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EXPERIMENT - 6

Snort IDS/IPS Using TryHackMe

Aim :

To understand and demonstrate the working of Snort as an Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) by detecting real-time network threats using the TryHackMe platform.

Procedure:

Step 1: Login to Platform

Open TryHackMe in a browser and log in using your account.

Step 2: Open Snort Room

Search for a Snort-related room and join it to begin the experiment.

Step 3: Start AttackBox

Launch the AttackBox, which provides a ready-to-use Linux environment for performing the tasks.

Step 4: Verify Snort Installation

Open the terminal and confirm that Snort is installed and ready to use.

Step 5: Run Snort in IDS Mode

Start Snort in detection mode so that it monitors network traffic and generates alerts for suspicious activity.

Step 6: Detect Network Threats

Perform the tasks given in the room and observe how Snort identifies and reports potential threats in real time.

Step 7: Configure IPS Mode

Enable prevention mode to allow Snort to block malicious traffic based on predefined rules.

Step 8: Analyze Alerts

Review the alerts generated by Snort and identify different types of network attacks.

Output:

tryhackme.com/room/snort

Room progress (17%)

contributors, and the Cisco Talos team.

Capabilities of Snort:

Snort has the following capabilities:

- Live traffic analysis
- Attack and probe detection
- Packet logging
- Protocol analysis
- Real-time alerting
- Modules & plugins
- Pre-processors
- Cross-platform support! (Linux & Windows)

Snort Modes

Snort has the following three main use modes:

- **Sniffer Mode**: Read and prompt IP packets in the console application.
- **Packet Logger Mode**: Log all IP packets (inbound and outbound) that visit the network.
- **NIDS (Network Intrusion Detection System) and NIPS (Network Intrusion Prevention System) Modes**: Log/drop the packets deemed malicious according to the user-defined rules.

Answer the questions below

Which IDS or IPS type can help you stop the threats on a local machine?

HIPS — ✓ Correct Answer

Which IDS or IPS type can help you detect threats on a local network?

NIDS — ✓ Correct Answer

Which IDS or IPS type can help you detect the threats on a local machine?

HIDS — ✓ Correct Answer

Which IDS or IPS type can help you stop the threats on a local network?

NIPS — ✓ Correct Answer

Which described solution works by detecting anomalies in the network?

NBA — ✓ Correct Answer

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Task 4 First Interaction with Snort

The First Interaction with Snort

First, let's verify Snort is installed. The following command will show you the instance version.

version check

```
user@ubuntu$ snort -V

   ,,_     -*> Snort! <*-
  o"  )~   Version 2.9.7.0 GRE (Build XXXXXX)
   ''''    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
           Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using libpcap version 1.9.1 (with TPACKET_V3)
           Using PCRE version: 8.39 2016-06-14
           Using ZLIB version: 1.2.11
```

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-V / --version	This parameter provides information about your instance version.
-c	Identifying the configuration file
-T	Snort's self-test parameter allows you to test your setup with this parameter.
-q	Quiet mode prevents Snort from displaying the default banner and initial information about your setup.

That was easy; let's move on to the next task and continue exploring snort modes.

Answer the questions below

Run the Snort instance and check the build number.

149 ✓ Correct Answer

Test the current instance with "/etc/snort/snort.conf" file and check how many rules are loaded with the current build.

4151 ✓ Correct Answer

Test the current instance with "/etc/snort/snortv2.conf" file and check how many rules are loaded with the current build.

1 ✓ Correct Answer

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Room progress (29%)

```
12/01-21:07:57.624205 216.58.214.142 -> 192.168.175.129
ICMP TTL:128 TOS:0x0 ID:63394 IpLen:20 DgmLen:84
Type:0  Code:0  ID:15  Seq:1  ECHO REPLY
0x0000: 00 0C 29 A5 B7 A2 00 50 56 E1 9B 9D 08 00 45 00  ..)....PV.....E.
0x0010: 00 54 F7 A2 00 00 80 01 24 13 D8 3A D6 8E C0 A8  .T......$..:....
0x0020: AF 81 00 00 BE B6 00 0F 00 01 2D E4 A7 61 00 00  ..........-..a..
0x0030: 00 00 A4 20 09 00 00 00 00 00 10 11 12 13 14 15  ... ............
0x0040: 16 17 18 19 1A 1B 1C 1D 1E 1F 20 21 22 23 24 25  .......... !"#$%
0x0050: 26 27 28 29 2A 2B 2C 2D 2E 2F 30 31 32 33 34 35  &'()*+,-./012345
0x0060: 36 37                                            67

=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+=+
```

Note that you can use the parameters both in combined and separated form as follows;

- snort -v
- snort -vd
- snort -de
- snort -v -d -e
- snort -X

Make sure you understand and practice each parameter with different types of traffic and discover your favourite combination.

Answer the questions below

You can practice the parameter combinations by using the traffic-generator script.

No answer needed ✓ Correct Answer

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Room progress (35%)

Activate the Data Acquisition (DAQ) modules and use the afpacket module to use Snort as an IPS; `-i eth0:eth1`

Identifying interfaces, note that Snort IPS requires at least two interfaces to work. Now run the traffic-generator script **as sudo** and start **ICMP/HTTP traffic**.

running IPS mode

```
user@ubuntu$ sudo snort -c /etc/snort/snort.conf -q -Q --daq afpacket -i eth0:eth1 -A console
Running in IPS mode

12/18-07:40:01.527100  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.175.131 -> 192.168.175.2
12/18-07:40:01.552811  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 172.217.169.142 -> 192.168.1.18
12/18-07:40:01.566232  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.175.131 -> 192.168.175.2
12/18-07:40:02.517903  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 192.168.1.18 -> 172.217.169.142
12/18-07:40:02.550844  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 172.217.169.142 -> 192.168.1.18
^C*** Caught Int-Signal
```

As you can see in the picture above, Snort blocked the packets this time. **We used the same rule with a different action (drop/reject).** Remember, for the scope of this task, our point is the operating mode, not the rule.

Answer the questions below

Investigate the traffic with the default configuration file.

`sudo snort -c /etc/snort/snort.conf -A full -l .`

Execute the traffic generator script and choose **"TASK-7 Exercise"**. Wait until the traffic stops, then stop the Snort instance. Now analyse the output summary and answer the question.

`sudo ./traffic-generator.sh`

What is the number of the detected HTTP GET methods?

2 ✓ Correct Answer

You can practice the rest of the parameters by using the traffic-generator script.

No answer needed ✓ Correct Answer

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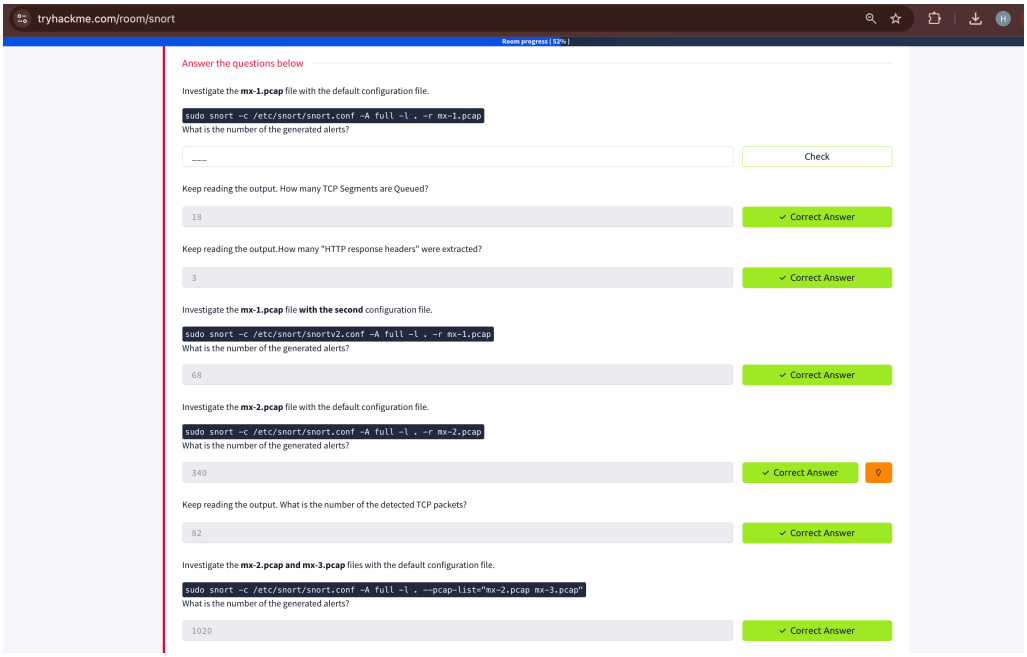

Result

Snort was successfully used as an IDS/IPS to detect and prevent network threats, demonstrating real-time network security monitoring in a controlled environment.